# MILITARY STANDARD

# FUZE, ARMY, DESIGN SAFETY

# CRITERIA FOR

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DEPARTMENT OF DEFENSE
WASHINGTON, D. C. 20301

Design, Safety, Army, Fuze, Criteria for MIL-STD-332 (MU).

1. This Military Standard is mandatory for use by all Departments and Agencies of the Department of Army.

2. Recommended corrections, additions or deletions should be addressed to the Commanding General, United States Army Munitions Command, ATTN: AMSMU-RE-F, Dover, New Jersey 07801.



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FOREWORD

1. The purpose of this standard is to establish specific requirements and objectives upon the design of fuzes (to include safing and arming devices) intended for use with Army munitions. It is not within the scope of this document to provide comprehensive design information to the fuze designer. However, sufficient discussion of design criteria and safing philosophy is included to enable those knowledgeable in this field to more clearly understand the intent of each requirement and objective.

2. It is recognized that it may be difficult to meet all the requirements of this standard for all Army Munitions. However, requiring documented recognition that a particular fuze does not conform to the requirements of this standard will serve as a continuing stimulus for achieving technological advances needed to meet all requirements.



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1. SCOPE

1.1 This standard describes and establishes the fundamental design safety criteria for fuzes (to include safety and arming (S&A) devices) used by the Army for the following munitions. (Separate criteria have been or will be issued to apply to nuclear munitions; all types of grenades; and munitions dispensed or emplaced manually such as mines, flares, signals, markers, demolition items, and other special warfare items.)"

a. Guided Missile Warheads

b. Rocket Warheads

c. Projectiles

d. Bombs

e. Bomblets

f. Destruct Systems

g. BW/CW Dispersal Systems

2. REFERENCED DOCUMENTS

2.1 The issues of the following documents in effect on the date of invitation for bids forms a part of this standard to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-T-00339 - Tetryl
MIL-R-398 - RDX
MIL-R-21723 - RDX Composition CH-6
MIL-E-14970 - Explosive Composition A5
MIL-E-81111 - PBXN-5
MIL-P-45486 - RDX Pellets
MIL-P-46464 - Pellets, Tetryl

STANDARDS

MILITARY

MIL-STD-331 - Test No. 115 - Static Detonator Safety



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MIL-STD-444 - Nomenclature and Definitions in the Ammunition Area
MIL-STD-1314 - (Navy) Safety Precautions for Explosive Loaded Items

OTHER PUBLICATIONS

WS-4660 - Dipam Explosive
WS-5003 - HNS, Type 1 or II, Grade A
USAMUCOM Reg. No. 705-11 R&D of Material, Fuze Design
AMC Reg. 385-224

(Copies of specifications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

## 3. DEFINITIONS

For interpretation of this standard, the following specific definitions apply.

3.1 Fuze.-Those components or mechanisms of a munition which act together to detonate the munition under the required circumstances and/or prevent unintended detonation of the munition.

3.2 Fuze system.-A complete system including, fuze, fuze setter, system interfaces (electrical and mechanical), and any other ancillary devices necessary for fuze operation.

3.3 Fuze Safety System.-A system consisting of one or more of the following:

3.3.1 Environmental sensors

3.3.2 Launch event sensors

3.3.3 Command functioned devices

3.3.4 A logic network which prohibits functioning of the fuze through all environments during transportation, storage, handling, installation, and launching of the munition.

3.4 Safe Fuze Design.-A fuze design which limits the probability of death or serious injury to friendly troops or property damage to an acceptable level. The generalized requirement to encompass all types of fuzes should not exceed one failure in $2.5 \times 10^6$.



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3.5 Safety Failure.-A failure of the fuze system prior to intended detonation which makes the fuze dangerous to handle., launch or results in a hazardous initiation of one of its explosive components.

3.6 Safe Separation Distance.-A distance between launcher and munition, following launch, beyond which an acceptably low probability or injury to launch personnel or damage to the launcher exists, if the munition should function.

3.7 Fail Safe Feature.-A characteristic of a fuze system or part thereof which is designed to result in a dud round when critical safing feature(s) malfunction.

3.8 Safety Feature.-An element or combination of elements designed to function if, and only if, an environmental force unique to a proper munition launch and/or post launch cycle is present.

3.9 Independent Safety Feature.-Each of several features is defined as independent if the malfunction of one does not affect the integrity of the other(s).

3.10 Fuze Arming.-A fuze is armed when it permits functioning of the munition as intended.

3.11 Arming Delay.-A time delay between launch initiation and fuze arming employed to obtain safe separation distance.

3.12 Environment.-The total set of physical conditions to which the fuze is exposed.

3.13 Pre-launch.-All conditions prior to the launch cycle.

3.14 Launch cycle.-The period between the time the munition is irreversibly committed to launch and the time the munition leaves the launcher.

3.15 Post-Launch.-Period of time subsequent to launch.

3.16 Environmental Force.-A stimulus or energy release obtained from the environment.

3.17 Explosive Train.-An arrangement of a series of pyrotechnical and/or explosive elements arranged to initiate the munition.



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3.18 Sensor, Sensing Elements.-A component or series of components designed to react to an environmental force.

3.19 Safe and Arming Device.-A mechanism which prevents an unintended detonation and in turn allows the explosive train of the munition to function.

## 4. GENERAL REQUIREMENTS

4.1 Applicability.-The provisions of this standard apply to:

4.1.1 All fuze systems in (1) exploratory development (6.21) where work has progressed to the point of assignment of an XM model designator to the item; (2) advanced development (6.31); (3) engineering and operational systems development (6.41) and (6.71); and (4) advance production engineering (APE).

4.1.2 This standard does not apply to fuzes which were type classified Standard A prior to the effective date of this standard or the Missile Systems currently in ED phase or beyond.

4.2 Action.-All activities engaged in the research and development of fuzes for Army munitions within the scope of this instruction will take necessary action to implement this policy.

## 5. DETAILED REQUIREMENTS

5.1 The following features, procedures, or controls are requirements and will be incorporated into the design of all fuzes under the scope of this document unless specifically exempted.

5.1.2 Safety System

5.1.2.1 The safety system of fuzes will comprise at least two independent safing features, both of which are capable of preventing an unintended detonation of the munition. Each of these safing features will be actuated by at least one separate environmental force. One feature must provide safe separation. A launch signal can be used to initiate one of the safing features.

5.1.2.2 If the primary explosive materials, more sensitive than the lead and booster explosives, (see 5.1.4 below) are housed in the interrupter, a single interrupter locked by the two safing features referred to, is acceptable since an omission of the



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interrupter will not result in a safety failure. If the element containing the primary explosive is mounted "in-line" and if the omission of a single interrupter could lead to a safety failure, the design must include two interrupters.

5.1.3 Explosive Train Interruption

5.1.3.1 Explosive train interruption will be positioned so as to isolate primary explosives using at least one interrupter (barrier, shutter, slider, rotor) to separate this explosive from the lead and booster explosives before arming. The interrupter will have positive, mechanical locks.

5.1.3.2 When the explosive train does not contain primary explosives, (e.g. EBW), interruption, shielding, and other protection, the initiation system must be designed to provide at least the same degree of fuze safety obtained with an interrupted train employing primary explosives.

5.1.3.3 Effectiveness of the interruption will be determined by subjection to Test No. 115 of MIL-STD-331. As a minimum, the design must meet the criteria for passing this test.

5.1.4 Lead and Booster Explosives.-The following explosives are the only ones permitted in a position leading to the initiation of the main charge without interruption when the fuze is in the safe condition:

| | For Applications For: | | |
|---|---|---|---|
| | Navy | Air Force | Army |
| Tetryl MIL-T-00339 | Yes | Yes | Yes |
| CH6, MIL-R-21723 (Nord) | Yes | Yes | Yes |
| PBXN-S, MIL-E-81111 (WEP) | Yes | Yes | No |
| DIPAM, GrA, WS-4660-B | Yes | Yes | Yes |
| HNS, T1, or T2 GrA, WS5003C | Yes | Yes | Yes |
| RDX, MIL-R-398 | No | Yes | Yes |
| Expl. Comp. A5 - (MIL-E-14970) | No | Yes | Yes |
| RDX Pellets, MIL-P-45486 | No | Yes | Yes |
| Tetryl Pellets, MIL-P-46464 | Yes | Yes | Yes |



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5.1.4.1 Alternative Lead and Booster Explosives.-If it is desired to use explosives other than those listed in 5.1.4, specific written approval is required from the cognizant technical authority (for Army: Explosive Lab., Picatinny Arsenal). Designs which do not meet this requirement or do not have specific written approval must be presented to an evaluation board (for the Army: the MUCOM Fuze Selection Advisory Board) for a safety evaluation for consideration of a waiver.

5.2 Objectives.-The following characteristics, features and controls shall be the objectives in the design of fuzes.

5.2.1 Energy for Arming.-The fuze safety system should not incorporate a stored energy mechanism which is used to remove an iterrupter of the fuze explosive train.

5.2.2 System requirement fail safe features should be included based on their applicability to system requirements.

5.2.3 Safe-Arming Indication.-One or more of the following options selected or combined in the fuze design:

a. A positive means of determining safe or armed conditions before and after installation into the round.

b. A feature which prevents installation of an unarmed fuze into the round.

c. A feature which prevents accidentally assembling the fuze in the armed condition. This option shall not be used alone if arming the assembled fuze in tests is a normal procedure prior to issue.

5.2.4 For multi-stage missiles and rockets, arming mechanisms should utilize last stage initiation as part of the arming sequence.

5.2.5 For electrical/electronic fuzing the "guidelines for the evaluation for the safety of Army Nuclear Weapon System" should be utilized.

5.2.6 If the fuze mechanism does not receive the necessary arming energy at or after launch, it should remain in the safe position.



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5.2.7 Safety failure rate should not exceed 1 in 2.5 million, unless the consequences of a malfunction are negligible or conversely are so severe to warrant a more stringent requirement.

5.2.8 Quality Control.-The design of the fuze should facilitate the use of automatic inspection equipment for all characteristics which assure the safety and functioning of the fuze.

5.3 General Guidance.-New fuze designs should be guided by, but not restricted to, procedures and features which have proved effective and valuable in prior fuze developments. In the area of design analysis, two such procedures are, the pre-design analysis of potential hazards and the safety failure mode analysis.

5.3.1 Pre-design Analysis of Potential Hazards.-The purpose of a fuze safety system is to prevent accidental actuation of a munition during assembly, handling, transportation, and storage, and in operational use until the munition is a safe distance from the launcher. To assure maximum safety in these phases, the fuze should be designed to provide protection in abnormal environments, conditions and personnel actions which may occur, as well as in the environments, conditions and actions which are normal. It is less likely that a design safety consideration will be overlooked if as complete an analysis as possible is made of the abnormal environments, conditions, and personnel actions which may occur in the phases before safe separation and this analysis is used as a guide to prepare design requirements. Contributions should be made by any persons knowledgeable in the fields of transportation and use of the munitions. Sever accident situations, such as fire, crushing and striking a hard surface at high velocity, should be included but judgement and experience must be used to give preference to situations which are more likely to arise. The process of preparing this analysis, to express those hazardous conditions in engineering terms, (such as environment magnitudes and physical description of personnel actions), which are useful to designers in determining needed characteristics of safety devices, is referred to, as predesign analysis of potential hazards, this should be a normal thought process of every designer. Formalizing it in report form is an aid in assuring thoroughness and completeness.



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5.3.2 Safety Failure Mode and Effects Analysis.-A useful method for analyzing the safety of a fuze design is a safety failure mode and effects analysis considering potential accident and malfunction paths. This analysis is a systematic consideration of the effects on fuze safety of such things as ommision of parts, breaking of parts, malfunctioning of parts, out of order sequence, inspection procedures and personnel errors. It should consider abnormal as well as normal environments, conditions, and personnel actions. The series of events which can lead to an accident (the accident paths) are examined to determine that adequate precautions to preclude them have been taken in design or through inspection and administrative procedures. As in the potential hazards analysis, severe accident situations should be included and judgement and experience must be used to give more emphasis to situations which are more likely to occur. This analysis differes from the predesign analysis of potential hazards in that it is a study of the fuze safety on the basis of known or anticipated hardware characteristics and specifications intended to be incorporated in the hardware design.

5.3.3 The following guidance is given in regard to design features of the fuze safety system:

The basic fuze safety system should be designed to sense (and require for arming) conditions which are sufficiently unique to the normal munition launch and post launch cycles that the possibility of encountering the same condition at anytime other than during a proper launch is negligible.

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